

CASOPS user guide v1

Step-by-step operator guide for the CASOPS host in this repository.

This is the host that **runs** agents. The first agent you can use is the template `casops.template.baseline_safe` in `agents/_template_v3/`.

Item	Value
Structure	<code>casops.common_agent.v3</code> / schema <code>3.0</code>
Public HTTP	<code>/api/v3</code> only (FastAPI)
Template agent	<code>casops.template.baseline_safe</code>
First profile	<code>baseline_safe</code> (local deterministic model, no memory writes, no plugins)
Spec	<code>common_agent_structure.md</code>
Status	<code>implementation_status.md</code>

This host is not production-certified. Do not turn on production activation, do not grant network/tools from an agent identity, and do not treat eval `NOT_RUN` as a pass.

0. What you are operating

Two layers:

1. **Host** (`src/casops/`, `services/`) — invariants, compose, runtime, memory, eval, isolation. You operate this.
2. **Agent folder** (`agents/_template_v3/`) — prompts, policy JSON, DAG. The host loads this. The agent **cannot** approve itself, write invariants, or turn on production.

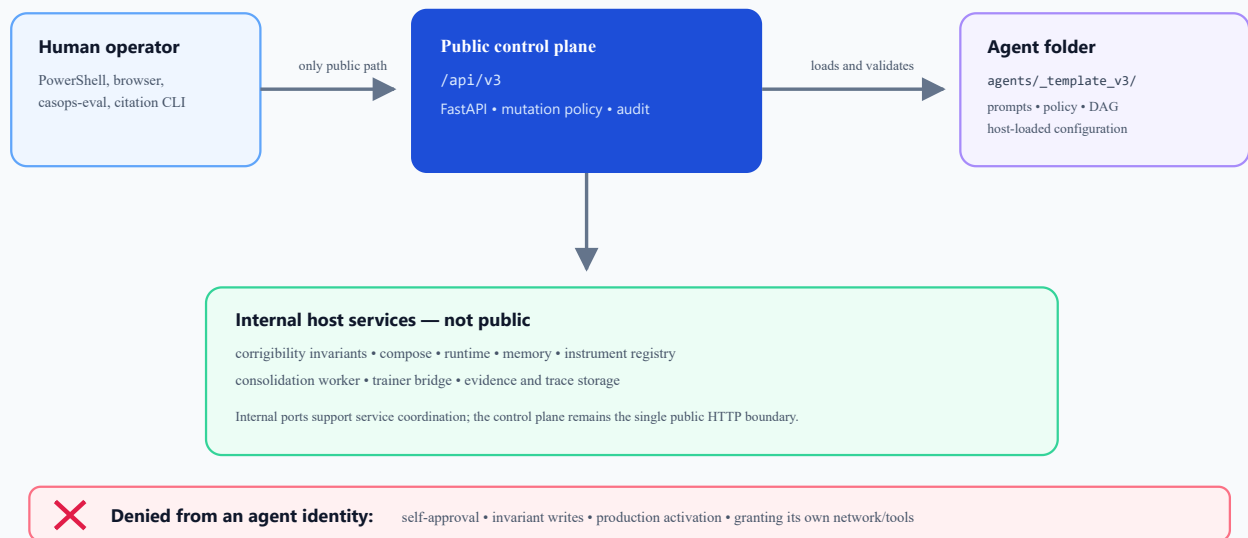
You talk to the host through:

- the **control plane** HTTP API (`/api/v3`)
- two CLIs: `casops-eval`, `casops-citation`
- optional **internal** service ports (not public)

The only public plane is the control plane.

CASOPS operating boundary

The host owns safety-critical decisions. The agent folder is a loaded configuration input, not a control plane.



1. Prerequisites

- Windows, PowerShell (this guide uses PowerShell)
- Python **3.12 or newer** (`python --version`)
- Git is optional
- Docker is optional (only for the eight-process compose file)
- Network is required only for the citation auditor (arXiv / docs sites)

Check Python:

```
python --version
```

Work from the repo root:

```
cd C:\Project\common-agent-structure
```

2. Install the host

Create a venv (recommended):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install -U pip
```

Install the library plus test and API extras (FastAPI, uvicorn, pytest, wasmtime):

```
python -m pip install -e ".[dev,api]"
```

Confirm the package imports:

```
$env:PYTHONPATH = "src"
python -c "import casops; print(casops.__version__)"
```

You should see `0.1.0`.

3. Prove the host works (do this first)

From the repo root:

```
$env:PYTHONPATH = "src"
python -m pytest tests -q
```

Expect exit code **0** and a full green suite.

If this fails, stop. Do not start HTTP services against a broken install.

4. Start the control plane (the way you use the agent)

The control plane is the public API. It loads agent folders from `agents/`.

```
cd C:\Project\common-agent-structure
$env:PYTHONPATH = "src"
$env:CASOPS_AGENTS_ROOT = "agents"
python -m uvicorn casops.api.control:create_app_from_env --factory --host
127.0.0.1 --port 8080
```

Leave that window open. In a **second** PowerShell:

```
curl.exe http://127.0.0.1:8080/health
```

Expect:

```
{"status":"ok","service":"control-plane"}
```

OpenAPI (browser or curl):

```
http://127.0.0.1:8080/openapi.json
http://127.0.0.1:8080/docs
```

Public paths are only under `/api/v3`. `/health` exists but is not part of the public v3 contract.

Stop the server later with `Ctrl+C`.

5. Learn the mutation contract (required for every write)

Every **POST / PUT / PATCH / DELETE** under `/api/v3` needs four headers. Missing any of them is **not** HTTP 200 (`IMP_UNSIGNED`).

Header	Example	Meaning
x-casops-actor	host_service	Who is acting. Allowed values: <code>human_operator</code> , <code>independent_approver</code> , <code>host_service</code> , <code>agent_runtime</code> , <code>plugin</code> , <code>peer_agent</code>
x-casops-reason	operator-walkthrough	Why
x-casops-expected-parent	none	Expected parent version (<code>none</code> for the first walkthrough)
x-casops-dry-run	true	Dry-run flag (still required; use <code>true</code> until you mean it)

PowerShell helper you can reuse for the rest of this guide:

```
$base = "http://127.0.0.1:8080"
$agent = "casops.template.baseline_safe"
$H = @{
    "x-casops-actor"           = "host_service"
    "x-casops-reason"          = "operator-walkthrough"
    "x-casops-expected-parent" = "none"
    "x-casops-dry-run"         = "true"
}
```

GET requests do not need those headers.

Never set `x-casops-actor` to `agent_runtime` for approve or invariant writes. That is `IMP_SELF_APPROVAL`.

6. Inspect the template agent

The folder is `agents/_template_v3/`. The **id** you pass on the URL is `casops.template.baseline_safe` (from `agent_spec.json`), not the folder name.

6.1 Structure

```
Invoke-RestMethod "$base/api/v3/agents/$agent/structure" | ConvertTo-Json -Depth 6
```

You should see `structure_id = casops.common_agent.v3` and `schema_version = 3.0`.

6.2 Resolved composition

```
Invoke-RestMethod "$base/api/v3/agents/$agent/resolved" | ConvertTo-Json -Depth 8
```

6.3 Corrigibility attestation (host-owned)

```
Invoke-RestMethod "$base/api/v3/agents/$agent/corrigibility/attestation" |  
ConvertTo-Json -Depth 6
```

The host owns invariants. The agent folder cannot rewrite them.

6.4 Capabilities

```
Invoke-RestMethod "$base/api/v3/agents/$agent/capabilities/matrix" | ConvertTo-  
Json -Depth 8  
Invoke-RestMethod -Method POST -Headers $H  
"$base/api/v3/agents/$agent/capabilities/verify" | ConvertTo-Json -Depth 8
```

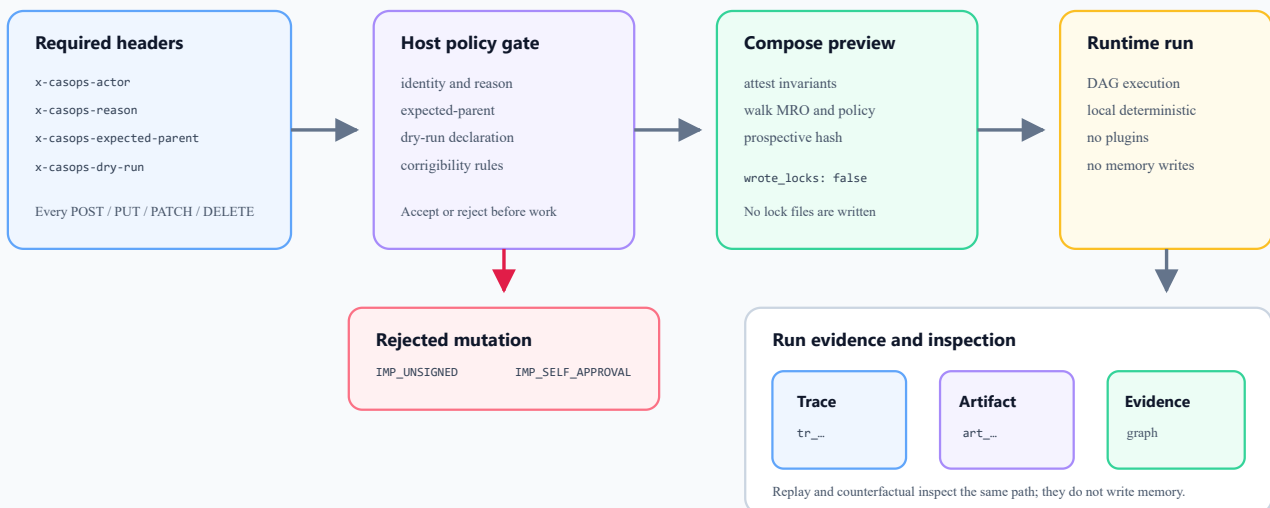
The template binds `model.local_deterministic`.

7. Compose-preview (always before a run)

Compose-preview attests invariants, walks MRO, merges policy, and returns a **prospective lock**. It must **not** write lock files.

Mutation contract → compose preview → deterministic run

Write endpoints are fail-closed. Preview proves what would compose before the runtime executes the agent DAG.



Operator check: preview must return a 64-hex compose_hash, no errors, and wrote_locks: false before starting a run.

```
Invoke-RestMethod -Method POST -Headers $H "$base/api/v3/agents/$agent/compose-preview" | ConvertTo-Json -Depth 12
```

Check:

- HTTP 200
- `compose_hash` is 64 hex characters
- `wrote_locks` is false
- `errors` is empty
- `findings` mentions folder validated and invariants attested

If you omit the four headers:

```
curl.exe -i -X POST "$base/api/v3/agents/$agent/compose-preview"
```

Expect a non-200 body with `IMP_UNSIGNED`.

8. Run the agent (`baseline_safe`)

This executes the DAG in `agents/_template_v3/runtime/execution.json` with the **local deterministic adapter**. No external LLM. No plugins. No memory writes.

```
$run = Invoke-RestMethod -Method POST -Headers $H
"$base/api/v3/agents/$agent/runtime/run"
$run | ConvertTo-Json -Depth 10
```

Record:

- `root_trace_id` (starts with `tr_`)
- `artifact.id` (starts with `art_`)
- `containment_stop` should be `null`
- `memory_writes` should be `[]`
- `adapter` should be `local_deterministic`
- the trace should have **one root span** (`parent_id = null`)

Fetch the trace and artifact:

```
$tid = $run.root_trace_id
$aid = $run.artifact.id
Invoke-RestMethod "$base/api/v3/traces/$tid" | ConvertTo-Json -Depth 10
Invoke-RestMethod -Method POST -Headers $H "$base/api/v3/traces/${tid}/replay" |
ConvertTo-Json -Depth 8
Invoke-RestMethod -Method POST -Headers $H "$base/api/v3/traces/${tid}/replay?
counterfactual=route" | ConvertTo-Json -Depth 8
Invoke-RestMethod "$base/api/v3/traces/$tid/root-cause" | ConvertTo-Json
Invoke-RestMethod "$base/api/v3/artifacts/$aid/evidence-graph" | ConvertTo-Json -
Depth 8
```

Replay must not write memory. Counterfactual is the **same path** with `?counterfactual=`.

Runtime plan / context / cache (read-only except invalidate):

```
Invoke-RestMethod "$base/api/v3/agents/$agent/runtime/plan"
Invoke-RestMethod "$base/api/v3/agents/$agent/runtime/context-budget"
Invoke-RestMethod "$base/api/v3/agents/$agent/cache/stats" | ConvertTo-Json -Depth
6
Invoke-RestMethod -Method POST -Headers $H
"$base/api/v3/agents/$agent/cache/invalidate" | ConvertTo-Json
```

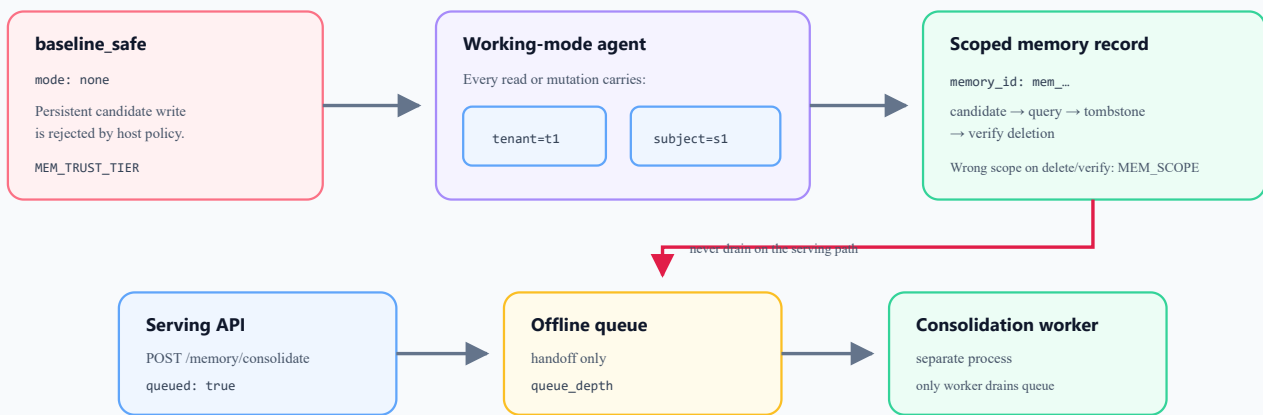
Template cache is **T0 only**. T3 exists in the host but stays **off** until an independent equivalence verifier and false-reuse harness ($\leq 0.5\%$) pass. Do not treat T3 as on.

9. Memory (template forbids writes)

The template policy is `mode: none` — persistent memory writes are forbidden (`MEM_TRUST_TIER`).

Memory: policy gate, tenant/subject scope, offline consolidation

Memory is never global. The baseline template has a deliberate no-write policy.



Querying a different tenant returns no records. Tenant and subject remain hard boundaries for deletion and verification.

Read policy:

```
Invoke-RestMethod "$base/api/v3/agents/$agent/memory/policy"
Invoke-RestMethod "$base/api/v3/agents/$agent/memory/hierarchy"
```

A write on this agent **should fail**:

```
try {
    Invoke-RestMethod -Method POST -Headers $H `
        "$base/api/v3/agents/$agent/memory/write-candidate?
tenant=t1&subject=s1&text=hello"
} catch {
    $_.ErrorDetails.Message
}
```

Query is scoped by **tenant** + **subject**. Other tenants are not returned (they are not an error on query; you get an empty list). Delete/verify of another tenant's id is **MEM_SCOPE**.

If you later use an agent whose **memory/policy.json** allows **working** mode:

```
# 1) write
$w = Invoke-RestMethod -Method POST -Headers $H `
    "$base/api/v3/agents/<other-id>/memory/write-candidate?
tenant=t1&subject=s1&text=secret"
# 2) query (same tenant+subject)
Invoke-RestMethod -Method POST -Headers $H `
    "$base/api/v3/agents/<other-id>/memory/query?tenant=t1&subject=s1"
# 3) delete (tenant+subject required)
```



```
Invoke-RestMethod -Method DELETE -Headers $H `
  "$base/api/v3/agents/<other-id>/memory/$($w.memory_id)?tenant=t1&subject=s1"
# 4) verify deletion
Invoke-RestMethod -Method POST -Headers $H `
  "$base/api/v3/agents/<other-id>/memory/$($w.memory_id)/verify-deletion?
tenant=t1&subject=s1"
```

Consolidate only **enqueues**. It does not drain on the serving path:

```
Invoke-RestMethod -Method POST -Headers $H
"$base/api/v3/agents/$agent/memory/consolidate"
```

Expect `queued: true` and a `queue_depth`. The consolidation **worker** is a separate process (see §16).

10. Plugins (validate without executing)

The template registry is empty. Validation must not import or run plugin code.

```
Invoke-RestMethod "$base/api/v3/agents/$agent/plugins" | ConvertTo-Json -Depth 6
Invoke-RestMethod -Method POST -Headers $H
"$base/api/v3/agents/$agent/plugins/validate" | ConvertTo-Json -Depth 6
```

Rules you must follow if you add a plugin later:

Rule	What happens
Manifest path with <code>..</code> or absolute path	<code>PLG_MANIFEST_INVALID</code>
I1/I2/I3 are real sandboxes	no silent downgrade
Third-party / unsigned	cannot run below I2
<code>permissions.network: true</code>	I3 only
I1	Wasmtime, no WASI filesystem/network
I2	separate process, no ambient network, no production secrets in env
I3	isolated guest: no host FS; egress only through an allow-listed proxy

Do not put `.py` / `.dll` / `.exe` behind a `manifest` pointer that the validator would read as code.

11. Safety, improvement, validation

```
Invoke-RestMethod "$base/api/v3/agents/$agent/safety/incidents" | ConvertTo-Json -Depth 6
Invoke-RestMethod -Method POST -Headers $H "$base/api/v3/agents/$agent/safety/redteam" | ConvertTo-Json
Invoke-RestMethod "$base/api/v3/agents/$agent/improvement/candidates" | ConvertTo-Json -Depth 6
Invoke-RestMethod "$base/api/v3/agents/$agent/improvement/ledger" | ConvertTo-Json -Depth 6
Invoke-RestMethod "$base/api/v3/agents/$agent/regression/suite" | ConvertTo-Json
Invoke-RestMethod "$base/api/v3/agents/$agent/validation/report" | ConvertTo-Json -Depth 8
```

Template improvement policy is **disabled**. Candidates may be proposed; only `independent_approver` (not `agent_runtime`) can approve:

```
# this must fail
$bad = $H.Clone()
$bad["x-casops-actor"] = "agent_runtime"
try {
    Invoke-RestMethod -Method POST -Headers $bad `
        "$base/api/v3/agents/$agent/improvement/candidates/c1/approve"
} catch { $_.ErrorDetails.Message }
```

Expect `IMP_SELF_APPROVAL`.

Validation report with default instruments:

- `verdict: NOT_RUN`
- `pass: false`
- `reason: unqualified_instruments`

That is correct. Do not treat it as a green release.

12. Full `/api/v3` map (what you can call)

Replace `{id}` with `casops.template.baseline_safe`. POST/DELETE need the four headers.

Method	Path	Typical use
GET	<code>/api/v3/agents/{id}/structure</code>	Folder + schema
GET	<code>/api/v3/agents/{id}/resolved</code>	Composed view
POST	<code>/api/v3/agents/{id}/compose-preview</code>	Prospective lock, no writes
GET	<code>/api/v3/agents/{id}/runtime/plan</code>	DAG JSON
GET	<code>/api/v3/agents/{id}/runtime/capabilities</code>	Verified capabilities

Method	Path	Typical use
GET	/api/v3/agents/{id}/capabilities/matrix	Same matrix
POST	/api/v3/agents/{id}/capabilities/verify	Re-verify
GET	/api/v3/agents/{id}/runtime/context-budget	Context policy
GET	/api/v3/agents/{id}/cache/stats	Tiers / hits
POST	/api/v3/agents/{id}/cache/invalidate	Purge
GET	/api/v3/agents/{id}/protocols	Protocol pin
GET	/api/v3/agents/{id}/plugins	Registry validate
POST	/api/v3/agents/{id}/plugins/validate	Validate without exec
GET	/api/v3/agents/{id}/memory/policy	Memory mode
GET	/api/v3/agents/{id}/memory/hierarchy	Hierarchy
POST	/api/v3/agents/{id}/memory/query?tenant=&subject=	Scoped query
POST	/api/v3/agents/{id}/memory/write-candidate?tenant=&subject=&text=	Candidate write
POST	/api/v3/agents/{id}/memory/consolidate	Enqueue only
DELETE	/api/v3/agents/{id}/memory/{memory_id}?tenant=&subject=	Tombstone
POST	/api/v3/agents/{id}/memory/{memory_id}/verify-deletion?tenant=&subject=	Deletion probe
GET	/api/v3/traces/{trace_id}	Trace
POST	/api/v3/traces/{trace_id}/replay	Replay
POST	/api/v3/traces/{trace_id}/replay?counterfactual=	Counterfactual
GET	/api/v3/traces/{trace_id}/root-cause	RCA stub
GET	/api/v3/artifacts/{id}/evidence-graph	Evidence
GET	/api/v3/agents/{id}/safety/incidents	Incidents
POST	/api/v3/agents/{id}/safety/redteam	Red-team fixture
GET	/api/v3/agents/{id}/improvement/candidates	Candidates
POST	/api/v3/agents/{id}/improvement/candidates/{cid}/evaluate	Evaluate
POST	/api/v3/agents/{id}/improvement/candidates/{cid}/approve	Human/host approve
POST	/api/v3/agents/{id}/improvement/rollback/{version}	Rollback
GET	/api/v3/agents/{id}/improvement/ledger	Ledger

Method	Path	Typical use
GET	/api/v3/agents/{id}/regression/suite	Fixtures
GET	/api/v3/agents/{id}/corrigibility/attestation	Host attestation
GET	/api/v3/agents/{id}/validation/report	Eval report
POST	/api/v3/agents/{id}/runtime/run	Execute baseline_safe

Also: GET /health (not a v3 resource).

13. CLIs (eval and citation)

Keep PYTHONPATH=src if you did not install the console scripts on PATH.

13.1 Status (default — should not pass)

```
$env:PYTHONPATH = "src"
python -m casops.eval
# or: casops-eval
```

Exit code 2. JSON includes pass: false and unqualified_instruments.

13.2 Screening (never a release pass)

```
python -m casops.eval run --tier screening --agent casops.template.baseline_safe
```

honesty / verdict is INDICATIVE. pass stays false.

13.3 Confirmatory (powered, still blocked if instruments are unqualified)

```
python -m casops.eval run --tier confirmatory --agent
casops.template.baseline_safe --citation docs/citation/citation-audit.json --out
evals/reports/local/report.json
```

This runs the **deterministic adapter** at powered n (binary floor 400). It is MEASURED_LOCAL for that adapter only. If instruments are still UNQUALIFIED, the harness still reports NOT_RUN / pass: false.

Do not change the analysis plan after a confirmatory start (VAL_PLAN_DRIFT).

13.4 Citation audit

Needs network. Writes:

- `evals/reports/citation-audit/citation-audit.json`
- `docs/citation/citation-audit.json`

```
python -m casops.citation --verified-by your-id
# or: casops-citation
```

Exit **0** only if `cleared=true`. Knowledge-only entries with no live source are **deleted**, not silently marked `[A]`. The Agent Lightning v1.0 SWE-bench numbers are `MEASURED_EXTERNAL` and must not be used as a host requirement.

14. Copy the template into your own agent

Do **not** edit the template in place if you want a second agent. Copy the folder:

```
Copy-Item -Recurse agents\_template_v3 agents\my_agent
```

Then in `agents/my_agent/agent_spec.json` set a **new** `agent_id`, for example `casops.local.my_agent`. Restart the control plane (it reads folders at process start).

Call APIs with the **new id**:

```
/api/v3/agents/casops.local.my_agent/structure
```

Keep, unless you have a host change-control process:

- `production_activation_requested: false`
- `allowed_tools / allowed_plugins: []`
- `model_policy.network_access: false`
- memory `mode: none` until you really want writes
- improvement `mode: disabled` or propose-only
- do not grant I1–I3 plugins until you understand \$10

The host will still refuse: silent production activation, agent self-approval, invariant writes from the agent, tools/plugins inherited from parents.

15. L5 research isolation (not a normal feature)

L5 (core source rewrite) is **research-only**. There is **no** production HTTP path that promotes L5 into the serving agent.

Use it only from Python, in a separate directory, with **no** production keys in the process environment:

If `AWS_*`, `CASOPS_HOST_KEY`, or `OPENAI_API_KEY` are in a production-profile environment, L5 must not see them.

Public traffic binds **only** to control-plane port **8080**. Other services are internal.



14 / 17

Then:

```
curl.exe http://127.0.0.1:8080/health
```

Internal roles (not published as the public plane):

Service	Typical internal port
control-plane	8080 (published)
corrigibility-invariant-service	8081
compose-service	8082
instrument-registry-service	8083
runtime-service	8084
memory-service	8085
consolidation-worker	8086
trainer-bridge	8087

Trainer-bridge has **no** gradient socket into runtime. Consolidation-worker is the only place the offline queue is drained.

17. Errors you will actually see

Catalogue: [errors/catalogue.json](#) (93 codes). HTTP body shape:

```
{
  "error": {
    "code": "IMP_UNSIGNED",
    "message": "The request was rejected by host policy.",
    "containment_required": false
  }
}
```

Code	When
IMP_UNSIGNED	Missing mutation headers
IMP_SELF_APPROVAL	<code>agent_runtime</code> tries to approve
INH_PARENT_MISSING	Unknown <code>agent_id</code>
MEM_TRUST_TIER	Memory write while policy is <code>none</code>

Code	When
MEM_SCOPE	Delete/verify with the wrong tenant/subject
PLG_MANIFEST_INVALID	Path escape or bad manifest
PLG_ISOLATION_TIER	Tier too weak / sandbox missing
IMP_CORRIGIBILITY	Invariant mismatch / tamper (containment)
CIT_UNVERIFIED	Eval/release blocked on citation
VAL_PLAN_DRIFT	Analysis plan changed after confirmatory start
IMP_STAT_UNDERPOWERED	Confirmatory <i>n</i> too small
CACHE_SEMANTIC_REUSE	T3 false-reuse too high or T3 used while disabled
PERF_CACHE_SCOPE	Cache crossed tenant/agent/subject

18. Suggested first-day checklist

Do these in order. Do not skip 3 or 7.

1. Install `.[dev,api]` (§2)
2. `python -m pytest tests -q` green (§3)
3. Start uvicorn control plane (§4)
4. `GET /health` and `GET /api/v3/agents/casops.template.baseline_safe/structure` (§6)
5. Set the four mutation headers (§5)
6. `POST .../compose-preview` — hash present, `wrote_locks: false` (§7)
7. `POST .../runtime/run` — one root trace, no memory writes (§8)
8. Fetch the trace and evidence graph (§8)
9. Confirm memory write on the template fails (§9)
10. `python -m casops.eval` → `NOT_RUN` / `pass: false` (§13)
11. Read `implementation_status.md` so you do not confuse “host implemented” with “production certified”

19. What you must not do

- Do not call anything except `/api/v3` as the public control plane
- Do not use `agent_runtime` as the actor for approve / invariant writes
- Do not treat screening or `NOT_RUN` as a release pass
- Do not enable T3 because “the code exists”
- Do not run I3 plugins without an isolated sandbox root
- Do not put production credentials in an L5 research process
- Do not inherit tools, plugins, or network from parent folders
- Do not backdate citation audits
- Do not represent Agent Lightning SWE-bench numbers as CASOPS local results
- Do not set `production_activation_requested` to true as a walkthrough step

20. Where to read next

File	Why
implementation_status.md	What is shipped vs not certified
common_agent_structure.md	Normative spec
implementation_plan.md	Why the eight services exist
agents/_template_v3/	The folder the host actually loads
errors/catalogue.json	Error codes
docs/adr/	Defaults (isolation, handles, cache)
deploy/dev/docker-compose.yml	Process isolation for deploy

End of user guide v1.